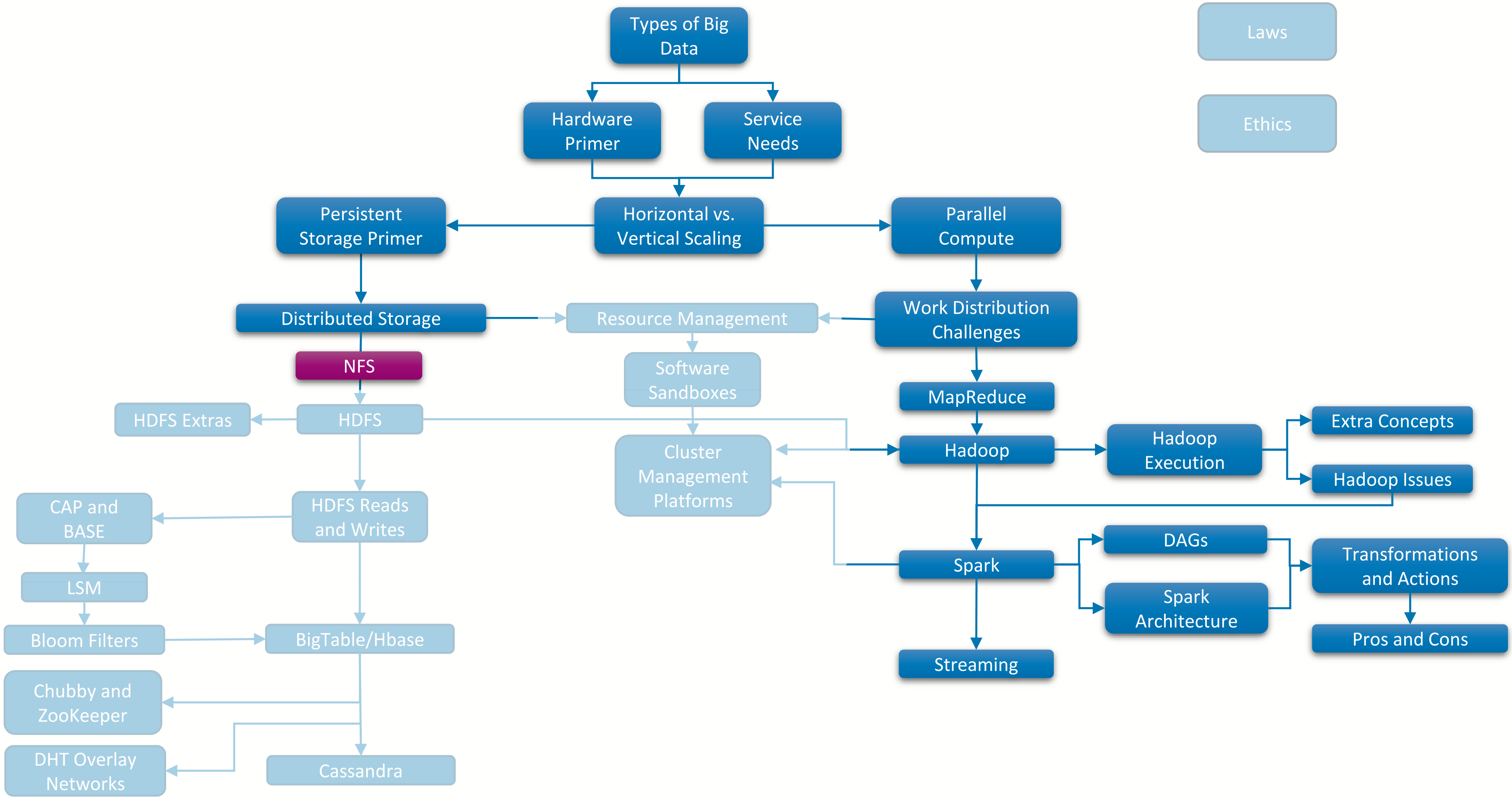
W5-03: Network File System

COMPSCI4064 (H) and COMPSCI5088 (M)

Dr. Richard McCreadie

Where Are We?



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Network File System (NFS)

- The Network File System (NFS) is a basic distributed file system originally developed back in 1984, and is still common today

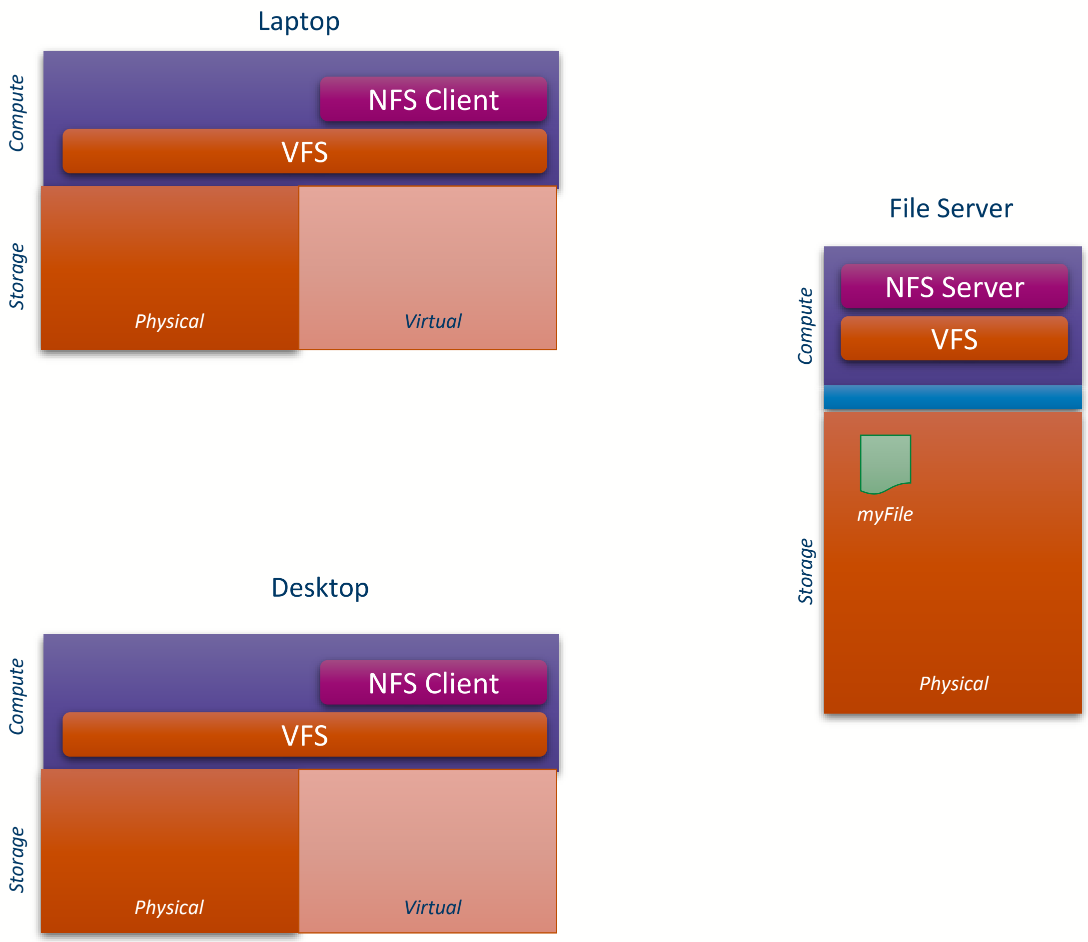
- It follows a client–server architecture
  - Servers host files
  - Clients can read and write those files

- NFS is on the server side both stateless and locking
  - Stateless: If the server control process no data is lost
  - Locking: Only one client can edit a file at once

NFS Architecture

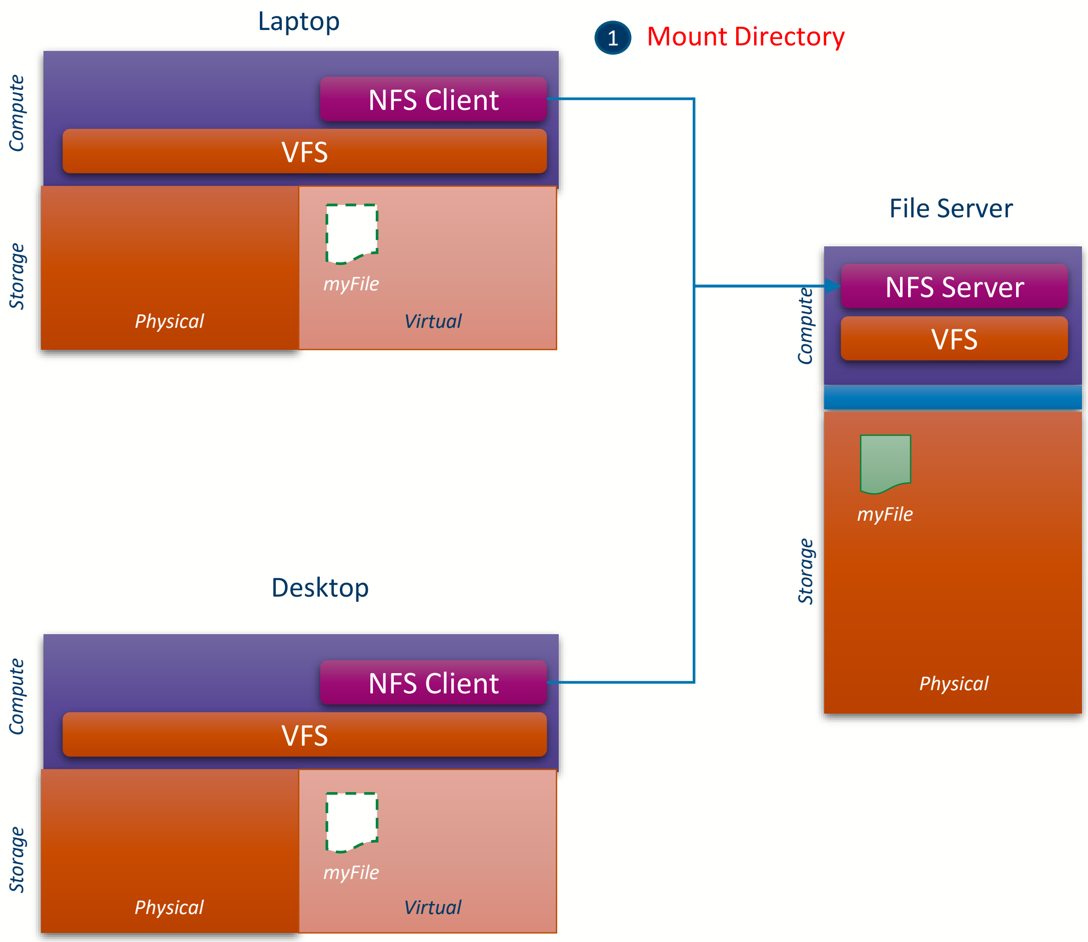
- Within most operating systems, there is a software file management layer
  - This allows devices of different formats to expose and read/write files via a standard API
  - In Linux this is the Virtual File System (VFS)

- The NFS Client uses this layer to mirror a file directory held on another machine
  - When a read request is made, a copy of the file is transferred over the network
  - When a write request is made, first a locking request is made, then writes can be made to the file, unlocking the file once done



Mounting

- The user can request that an NFS client mirror a specified remote directory exposed by an NFS server
  - This process is known as 'mounting'
- NFS Mounting is a multi-stage process
  - The client first requests the mount port number from the portmap service on the NFS server
  - The client pings the specified port on the server to make sure its live
  - The client then makes a mount request to the server, getting the file handle for the file system
  - The client connects to the port using the file handle and requests the information about the file system
- Multiple machines can mount the same file system
  - Parallel reads allowed, but not parallel writes



Advantages and Disadvantages

Advantages

- Simple (files exist in only one place)
- Widely compatible
- Low overhead
- (Relatively) fast
- Reliable

Disadvantages

- No native data replication, the server is a single point of failure
  - Often combined with RAID to provide a minimal level of defence against failures
- Inherently insecure – assumes a trusted network
- Sensitive to network congestion and server load

University of Glasgow

Course Coordinator
Dr. Richard McCreadie

Email: richard.mccreadie@glasgow.ac.uk
Room: SAWB 304

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